Hyperbolic Functions

Questions 1–70

Question 1. Simplify each of these, writing your answers in exponential form:

a) $2\sinh x$
b) $8\cosh x$
c) $4\sinh x - 4\cosh x$
d) $5\cosh x - 10\sinh x$

Question 2. Show that:

a) $2\sinh x + 4\cosh x \equiv e^{x}(3e^{-2x} + 1)$
b) $3\sinh x - 9\cosh x \equiv -3e^{-x}(e^{2x} + 2)$
c) $10\cosh x - 8\sinh x \equiv e^{-x}(e^{2x} + 9)$
d) $26\cosh x + 42\sinh x \equiv 2e^{x}(17e^{-2x} - 4)$

Question 3. Show that $55\sinh x - 35\cosh x$ can be written in the form $ae^{-x}(be^{2x} - 9)$, where a and b are constants to be found.

Question 4. By using the exponential forms of $\sinh x$ and $\cosh x$, find the value of each of these:

a) $\sinh 0$

b) $\cosh 0$

Question 5. Write down the exact value of each of these in terms of exponentials, and then to 3 significant figures:

a) $\sinh 1$
b) $\cosh 1$
c) $\sinh(-1)$
d) $\cosh(-1)$

Question 6. Show that $\sinh(-x) = -\sinh x$.

Question 7. Show that $\cosh(-x) = \cosh x$.

Question 8. By using the exponential forms of $\sinh x$ and $\cosh x$, find the value of each of these:

a) $\sinh(\ln 2)$
b) $\cosh(\ln 3)$
c) $\sinh(\ln(1\frac{1}{3}))$
d) $\cosh(\ln(1\frac{1}{4}))$

Question 9. Show that

$$\sinh(\ln k) = \frac{k^2 - 1}{2k}$$

where k is a positive constant.

Question 10. Show that

$$\cosh(2\ln k) = \frac{k^4 + 1}{2k^2}$$

where k is a positive constant.

Question 11. By using the exponential forms of $\sinh x$ and $\cosh x$, show that

$$\tanh x \equiv \frac{(e^x + 1)(e^x - 1)}{e^{2x} + 1}$$

Question 12. By using the exponential forms of $\sinh x$ and $\cosh x$, show that

$$\frac{1}{2}\left(\frac{1 + \tanh x}{1 - \tanh x}\right) = x$$

Question 13. Write down the exact value of each of these in terms of exponentials, and then to 3 significant figures:

a) $\tanh 0$

b) $\tanh 1$

c) $\tanh(-1)$

Question 14. Show that $\tanh(-x) = -\tanh x$.

Question 15. By using the exponential form of $\tanh x$, find the value of each of these:

a) $\tanh(\ln 3)$

b) $\tanh(-\ln 4)$

c) $\tanh(\ln \frac{1}{5})$

Question 16. Prove, for all real values of x, that $\sinh x + \cosh x > 0$.

Question 17. Prove, for all real values of x, that $\sinh x - \cosh x < 0$.

Question 18. Use exponentials to solve the following equations, writing your answers in exact form:

a) $\sinh x = 0$

b) $\sinh x = 1$

c) $\sinh x = -10$

Question 19. Use exponentials to solve the following equations, writing your answers in exact form:

a) $\cosh x = 1$

b) $\cosh x = 5$

c) $\cosh x = 23$

Question 20. Use exponentials to solve the following equations, writing your answers in exact form:

a) $\tanh x = 0$

b) $\tanh x = \frac{1}{3}$

c) $\tanh x = -\frac{4}{5}$

Question 21. Solve the following equations, writing your answers in exact form:

a) $\sinh x + \cosh x = 2$

b) $2\sinh x - \cosh x = 1$

c) $\sinh x - 2\cosh x = -2$

Question 22. Solve the following equations, writing your answers in exact form:

a) $2\cosh x - 3\sinh x = 1$

b) $5\sinh x - 6\cosh x = -7$

c) $3\cosh x - 5\sinh x = 2$

d) $4\sinh x + 3\cosh x = 4$

Question 23. Solve the following equations, writing your answers in exact form:

a) $2\sinh 2x = 2 - 7e^{2x}$

b) $3\sinh 2x = 2e^{2x} - 2$

Question 24.

a) Fully expand $(y^2 - 6y + 1)(y^2 + y + 1)$.

b) Use exponentials to solve $\cosh 2x = 2 + 5\cosh x$, writing your answers in exact form.

Question 25.

a) It is given that $f(y) = y^4 - 4y^3 + 4y - 1$.

(i) Show that $f(1) = 0$ and $f(-1) = 0$.

(ii) Fully factorise $f(y)$.

b) Use exponentials to solve $\sinh 2x = 4 \sinh x$, writing your answers in exact form.

Question 26. Solve the following equations, writing your answers in exact form:

a) $\sinh^2 x + \sinh x - 6 = 0$

b) $8 \cosh^2 x - x - 26 \cosh x + 15 = 0$

Question 27. Solve the following equations, writing your answers in exact form:

a) $\cosh^3 x - 5\cosh^2 x - 2\cosh x + 24 = 0$

b) $12\cosh^4 x - 44\cosh^2 x - 45 = 0$

Question 28. Solve the following equations, writing your answers in exact form:

a) $15\tanh^2 x - 8\tanh x + 1 = 0$

b) $12\tanh^2 x - 17\tanh x + 6 = 0$

Question 29.

a) It is given that $f(y) = y^3 - 7y^2 + 7y - 1$.

(i) Show that $f(1) = 0$.

(ii) Fully factorise $f(y)$.

b) Show that $x = 0$, $x = \ln(\sqrt{2} - 1)$ and $x = \ln(1 + \sqrt{2})$ are the real roots of the equation $3\tanh 2x = 4\tanh x$.

Question 30. Solve the following equations, writing your answers in exact form:

a) $\frac{1}{\cosh x} + \frac{1}{\sinh x} = e^{3x}$

b) $\frac{1}{\cosh x} + \frac{1}{\sinh x} = \frac{4}{e^x}$

c) $\frac{1}{\cosh x} + \sinh x = \frac{2}{e^x}$

d) $4\cosh x - \frac{1}{\sinh x} = \frac{2}{e^x}$

Question 31.

a) It is given that $f(y) = y^4 + 8y^3 + 2y^2 - 16y - 11$.

(i) Show that $f(-1) = 0$.

(ii) Given that $(y+1)^2$ is a factor of $f(y)$, fully factorise $f(y)$.

b) Use exponentials to solve

$$\frac{2\sinh x}{1+\cosh x} + \frac{1+\cosh x}{2\sinh x} = 3$$

writing your answers in exact form.

Question 32. Show that

$$\cosh^3 x + \sinh 3x = \frac{1}{8}(e^{3x} + 3e^x + 3e^{-x} - e^{-3x})$$

where a is a positive integer.

Question 33.

a) Show that

(i) $\sinh^3 x - 4\cosh^3 x = -(e^{-3x} + ae^x)$ where a is a positive integer.

(ii) $4\sinh^3 x + 4\cosh^3 x = e^{3x} + be^{-x}$ where b is a positive integer.

b) Hence show that

$$16\sinh^6 x - 16\cosh^6 x = -2(p\cosh 4x + q)$$

where p and q are positive integers.

Question 34. Prove $(\sinh x + \cosh x)^3 = e^{3x}$.

Question 35.

a) By using the exponential forms of $\cosh x$, prove that $4\cosh^3 x \equiv \cosh 3x + 3\cosh x$.

b) Hence show that the equation

$$4y^3 - 3y = \frac{5}{4}$$

has a real solution of the form $p^{1/4} + q^{1/4}$, where p is an integer and q and r are rational numbers.

c) Use this result to find the exact values of p, q, r.

Hyperbolic Identities

Question 36. Simplify the following expressions as far as possible, writing your answers in terms of $\sinh x$ and $\cosh x$ only:

a) $\cosh x \tanh x$

b) $\frac{\tanh x}{\sinh x}$

c) $\frac{\tanh x}{\cosh x}$

Question 37. Simplify the following expressions as far as possible, writing your answers in terms of $\sinh x$ and $\cosh x$ only:

a) $\sinh x \cosh x \tanh x$

b) $\frac{\sinh^2 x \cosh x \tanh^2 x}{\sinh x}$

c) $\frac{\sinh x}{\cosh^2 x \tanh x}$

d) $\frac{\tanh^2 x}{\sinh^2 x \cosh x}$

Question 38. By using the exponential forms of $\sinh x$ and $\cosh x$, prove that

$$\cosh^2 x - \sinh^2 x = 1$$

Question 39. By using the exponential forms of $\sinh x$ and $\cosh x$, prove that

$$\sinh 2x = 2\sinh x \cosh x$$

Question 40. By using the exponential forms of $\sinh x$ and $\cosh x$, prove that:

a) $\cosh 2x = \cosh^2 x + \sinh^2 x$

b) $\cosh 2x = 2\cosh^2 x - 1$

c) $\cosh 2x = 1 + 2\sinh^2 x$

Question 41. By using the exponential forms of $\sinh x$ and $\cosh x$, prove that

$$\tanh 2x = \frac{2\tanh x}{1+\tanh^2 x}$$

Question 42. Write each of these in terms of $\sinh x$ only:

a) $5\cosh^2 x + 3\sinh^2 x$

b) $2\sinh^2 x - 6\cosh^2 x$

Question 43. Write each of these in terms of $\cosh x$ only:

a) $9\cosh^2 x + 2\sinh^2 x$

b) $5\sinh^2 x - 8\cosh^2 x$

Question 44. Given that $\cosh 2x = 2\cosh^2 x - 1$ and $\cosh 2x = 1 + 2\sinh^2 x$, write each of these in terms of $\cosh 2x$ only:

a) $2\cosh^2 x + 5\sinh^2 x$

b) $8\sinh^2 x - 3\cosh^2 x$

Question 45. Use hyperbolic identities to solve the following equations, writing your answers in exact form:

a) $\cosh^2 x + \sinh^2 x = 3$

b) $2\cosh^2 x + 3\sinh x - 4 = 0$

c) $5\sinh x = 2\cosh^2 x$

d) $3\sinh^2 x = 2\cosh x + 5$

Question 46. Use hyperbolic identities to solve the following equations, writing your answers in exact form:

a) $\cosh 2x = 3 + 7\cosh x$

b) $5\cosh 2x + 8 = \sinh x - 7$

c) $8\cosh 2x = \cosh x + 9$

d) $\sinh x = 5 - 2\cosh 2x$

Question 47. Use hyperbolic identities to solve the following equations, writing your answers in exact form:

a) $\sinh 2x = 20 \sinh x$

b) $\frac{2}{3} \sinh 2x = \cosh x$

Question 48. Dylan is trying to solve the equation $3 \sinh x + \cosh^2 x = 5$. His working is as follows:

$$3 \sinh x + 1 - \sinh^2 x = 5$$

$$\Rightarrow 0 = \sinh^2 x - 3 \sinh x + 4$$

$$\Rightarrow 0 = (\sinh x - \frac{3}{2})^2 - \frac{9}{4} - 4$$

$$\Rightarrow 0 = (\sinh x - \frac{3}{2})^2 - \frac{25}{4}$$

$$\Rightarrow \sinh x = \frac{3}{2} \pm \frac{5}{2} = 4 \text{ or } -1$$

$$\Rightarrow x = \sinh^{-1} 4 \text{ or } \sinh^{-1}(-1)$$

a) Identify the two errors Dylan has made in his working.

b) Write out a correct method and find the exact values of x that solve the equation.

Question 49.

a) Prove $8\cosh^4 x \equiv \cosh 4x + 4\cosh 2x + 3$.

b) Solve the equation $2\cosh 4x - 23\cosh 2x - 33 = 0$.

Question 50. Solve the following equations, writing your answers in exact form:

a) $\sinh x(\sinh x + \cosh x) = 4$

b) $\cosh x(\sinh x + \cosh x) = 13$

c) $\sinh x(5\sinh x - 4\cosh x) = 3$

Question 51. Use hyperbolic identities to solve the following equations, writing your answers in exact form:

a) $3\tanh 2x - 4\tanh x = 0$

b) $10\tanh 2x = 9 + \tanh x$

Question 52. By using the exponential forms of $\sinh x$ and $\cosh x$, prove that

$$\sinh x \cosh y + \cosh x \sinh y \equiv \sinh(x + y)$$

Question 53. By using the exponential forms of $\sinh x$ and $\cosh x$, prove that

$$\cosh x \cosh y - \sinh x \sinh y \equiv \cosh(x - y)$$

Question 54. It is given that $\sinh(x + \ln 5) = \cosh x$.

a) Show that $\tanh x = -\frac{7}{13}$.

b) Hence show that $x = \frac{1}{2}(\ln 3 - \ln 2 - \ln 5)$.

Question 55. Use the exponential form of $\sinh x$ to prove

$$\sinh 3x = 4\sinh^3 x + 3\sinh x$$

Question 56. Use the exponential form of $\sinh x$ to prove

$$\cosh 3x = 4\cosh^3 x - 3\cosh x$$

Question 57. Use the identities proved in the previous two questions to solve the following equations:

a) $\sinh 3x = 15 \sinh x$

b) $\cosh 3x = 5 \cosh x$

Question 58. Given that $\sinh x = 3$, where $x > 0$, find the exact value of:

a) $\cosh x$

b) $\tanh x$

Question 59. Given that $\cosh x = 5$, where $x > 0$, find the exact value of:

a) $\sinh x$

b) $\tanh x$

Question 60. Given that $\tanh x = \frac{1}{3}$, where $x > 0$, find the exact value of:

a) $\sinh x$

b) $\cosh x$

Question 61. Find the value(s) of x that satisfy both of the following equations:

$$2\cosh x + \sinh x = 2$$

$$4\cosh^2 x - 11\sinh x = 4$$

Question 62. Find the value(s) of x that satisfy both of the following equations:

$$2\cosh^2 x - \sinh x = 8$$

$$2\cosh^2 x - 3\sinh x = 4$$

Question 63. Prove the following identities:

a) $(\sinh x + \cosh x)^2 \equiv \sinh 2x + \cosh 2x$

b) $(\sinh 2x - \cosh 2x)^2 \equiv \cosh 4x - \sinh 4x$

c) $1 - \tanh^2 x \equiv \frac{1}{\cosh^2 x}$

d) $\frac{1}{\tanh^2 x} - 1 \equiv \frac{1}{\sinh^2 x}$

Question 64. Prove

$$\sinh^3 x = \frac{1}{4}\sinh 3x - \frac{3}{4}\sinh x$$

Question 65. Prove

$$\cosh 3x - 3\cosh x = 2\cosh x(\cosh 2x - 2)$$

Question 66. Prove

$$\cosh 2x - 2\sinh^2 x \equiv 1$$

Question 67. Prove

$$3\sinh x - 7\sinh(\frac{x}{2}) = \sinh(\frac{x}{2})(6\cosh(\frac{x}{2}) - 7)$$

Question 68. Prove

$$\sinh x = 4\sinh(\frac{x}{4})\cosh(\frac{x}{4})\cosh(\frac{x}{2})$$

Question 69. Prove

$$\cosh^2(\frac{x}{3}) - 4\sinh^2(\frac{x}{6})\cosh^2(\frac{x}{6}) \equiv 1$$

Question 70. Prove

$$4\cosh(\frac{x}{2}) = 4\cosh^4(\frac{x}{4}) - \sinh^2(\frac{x}{2})\tanh^2(\frac{x}{4})$$